

Roll No. :

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: B. Tech ECE (IOT)

Semester: V

Name of the Course: Transducers, Actuators and Display Devices

Course Code: TEC591

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each question carries 10 marks

Q1	(10 Marks)	CO1
(a)	Define active and passive sensors. Give two examples of each and explain their working principles.	
	OR	
(b)	Differentiate between static and dynamic characteristics of a sensor with suitable examples.	
Q2	Write (10 Marks)	CO1
(a)	Explain the terms accuracy, offset, and linearity in the context of sensor static characteristics.	
	OR	
(b)	Explain the photoelectric effect and its application in sensor technology.	
Q3	(10 Marks)	CO1
(a)	Define the thermoelectric effect and discuss its application in temperature sensors.	
	OR	
(b)	Explain the piezoelectric sensor with industrial applications.	
Q4	(10 Marks)	CO2
(a)	Define a conductometric transducer. How does it work, and where is it commonly used?	
	OR	
(b)	Explain how a capacitive transducer works for displacement measurement. Derive an expression for capacitance in terms of plate area and separation.	
Q5	(10 Marks)	CO2
(a)	What is an interferometric optical transducer? Explain its principle using the concept of interference of light.	
	OR	
(b)	What is an electrochemical transducer? Give an example where this type of sensor is used.	

